

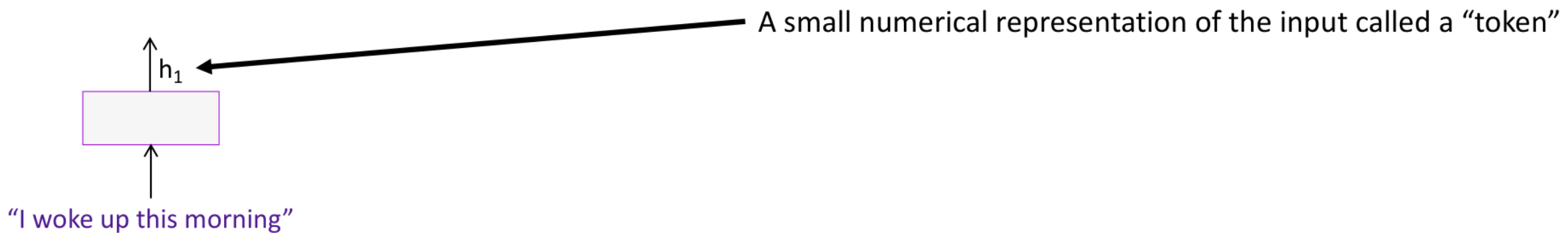
How Large Language Models Actually Work

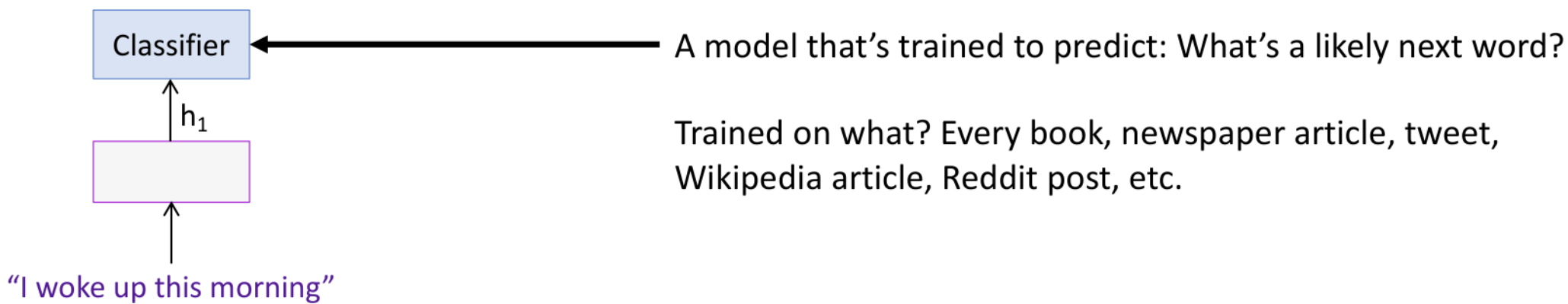
Robert Pless · Trustworthy AI · CSCI 6510 / EMSE 6992

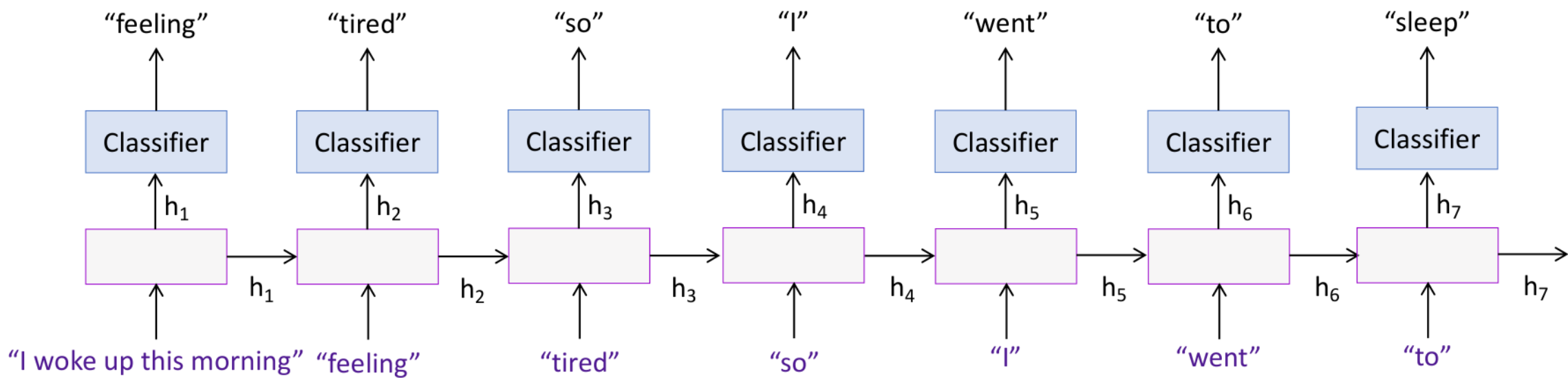
<https://jalammar.github.io/illustrated-word2vec/>

next-token prediction, + a lot of engineering

- Underneath ChatGPT, Claude, Gemini: a model that predicts the next chunk of text.
- The knowledge, “reasoning” and the helpful (?) personality is built on that one operation.
- And, so is everything that makes it hard to trust: confident wrong answers, different answers each time...

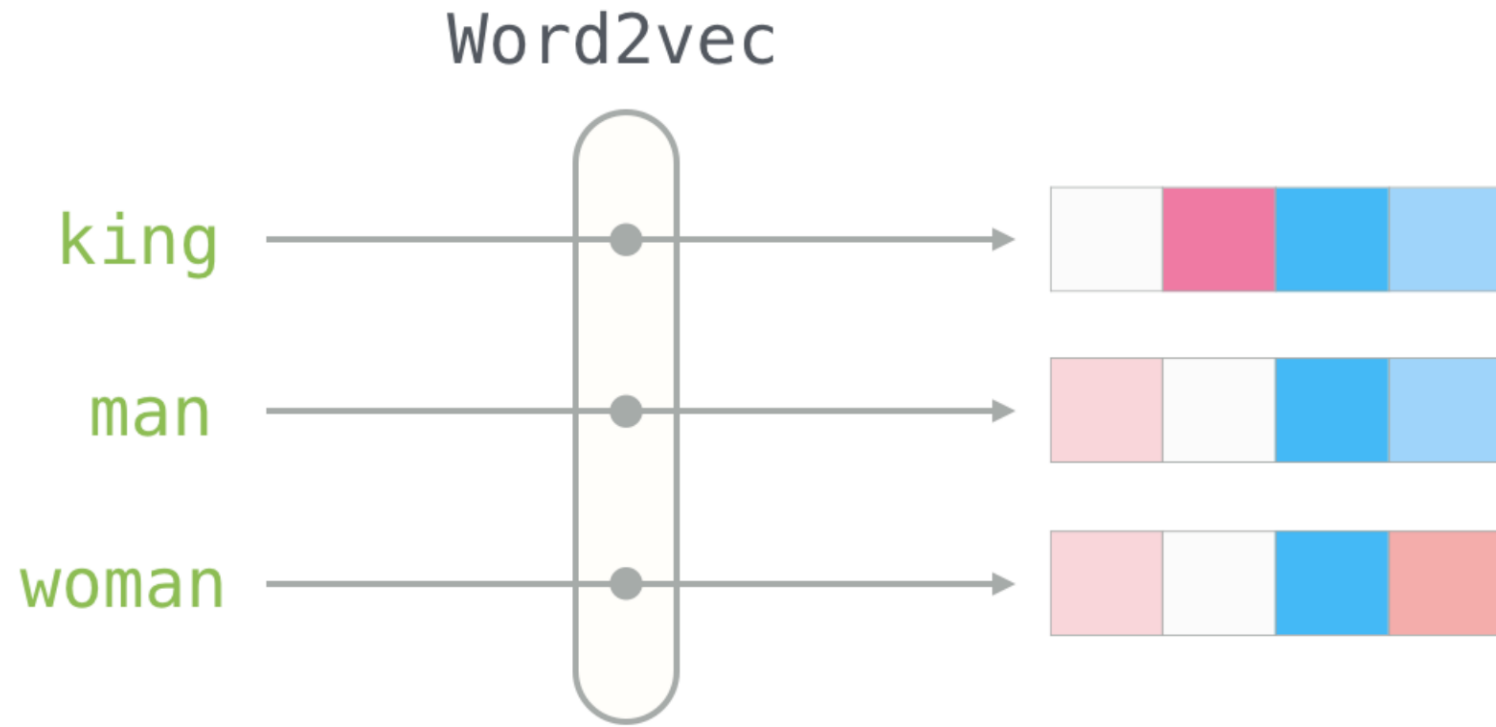






Step 1. What is a token?

- Going to start with what a token was a few years ago.



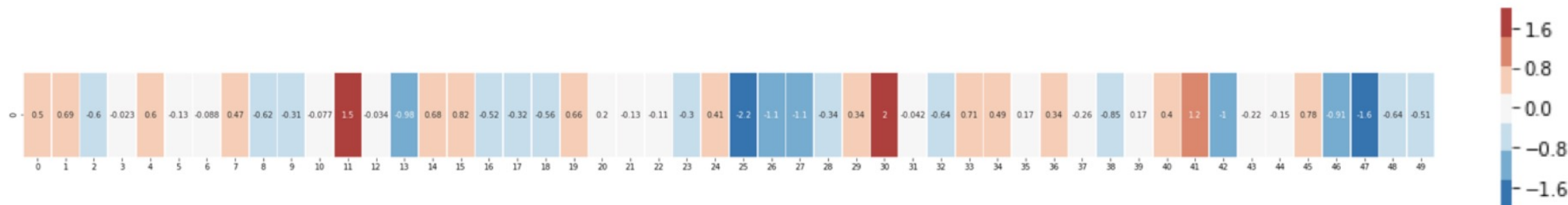
This is a word embedding for the word “king” (GloVe vector trained on Wikipedia):

```
[ 0.50451 , 0.68607 , -0.59517 , -0.022801, 0.60046 , -0.13498 , -0.08813 , 0.47377 , -0.61798 , -0.31012 , -0.076666, 1.493 , -0.034189, -0.98173 , 0.68229 , 0.81722 , -0.51874 , -0.31503 , -0.55809 , 0.66421 , 0.1961 , -0.13495 , -0.11476 , -0.30344 , 0.41177 , -2.223 , -1.0756 , -1.0783 , -0.34354 , 0.33505 , 1.9927 , -0.04234 , -0.64319 , 0.71125 , 0.49159 , 0.16754 , 0.34344 , -0.25663 , -0.8523 , 0.1661 , 0.40102 , 1.1685 , -1.0137 , -0.21585 , -0.15155 , 0.78321 , -0.91241 , -1.6106 , -0.64426 , -0.51042 ]
```

It's a list of 50 numbers. We can't tell much by looking at the values. But let's visualize it a bit so we can compare it to other word vectors. Let's put all these numbers in one row:



Let's color code the cells based on their values (red if they're close to 2, white if they're close to 0, blue if they're close to -2):



“king”



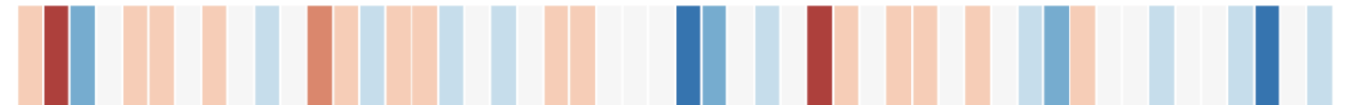
“Man”



“Woman”



queen



woman



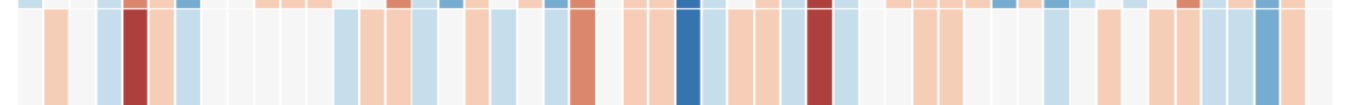
girl



boy



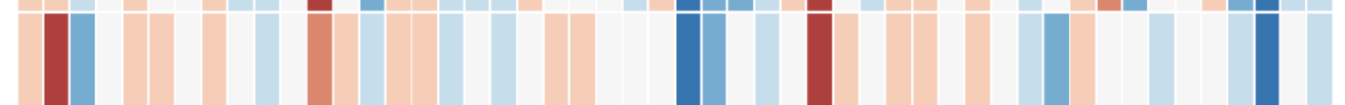
man



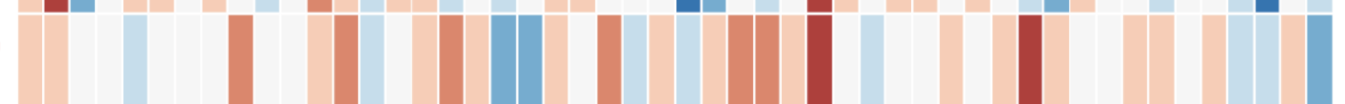
king



queen



water

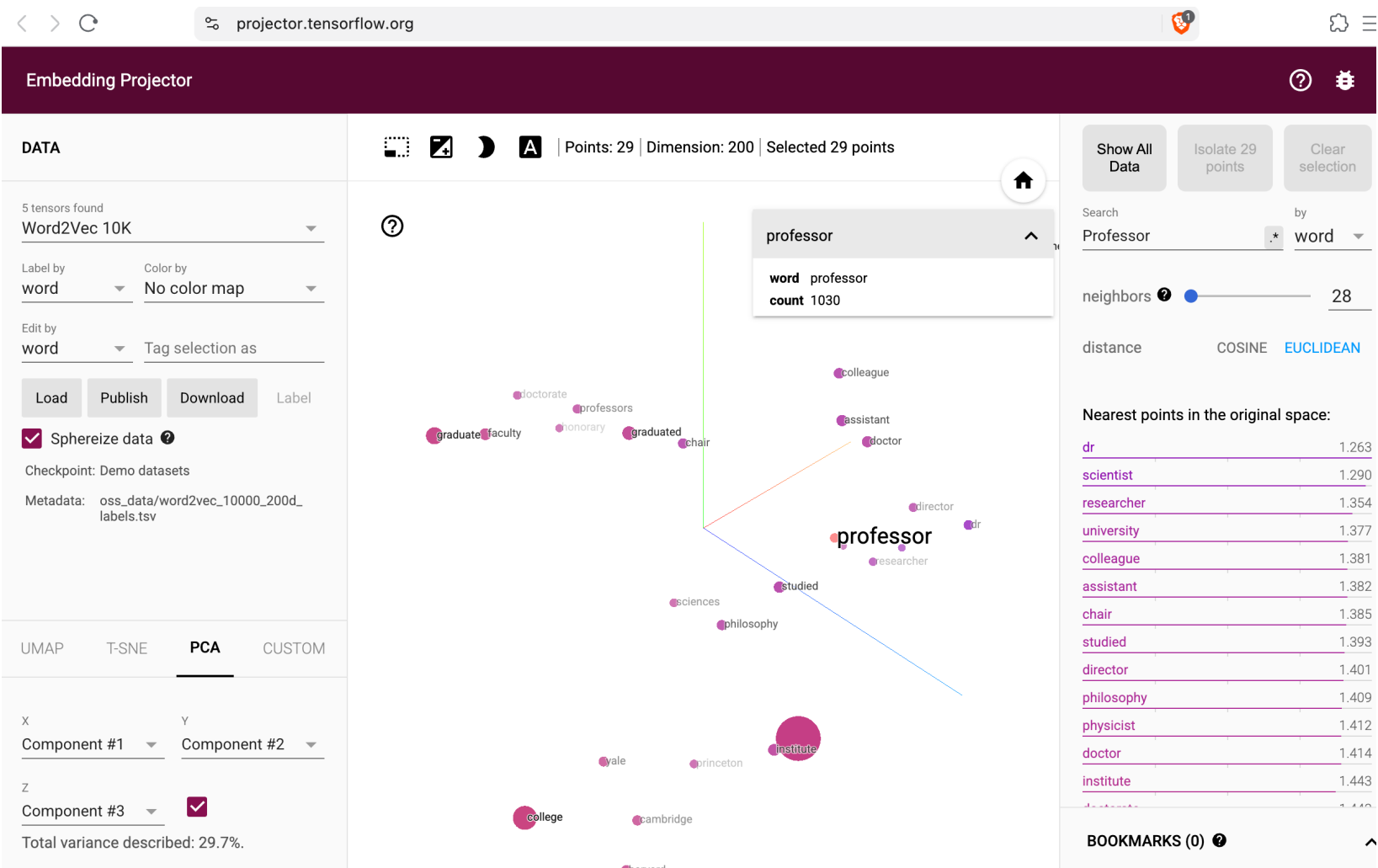


king - man + woman $\approx$ queen



The resulting vector from "king-man+woman" doesn't exactly equal "queen", but "queen" is the closest word to it from the 400,000 word embeddings we have in this collection.

https://projector.tensorflow.org/



Step 1. What is a token now?

Prompt builder

Build and edit your conversation messages.

User

X

This is a test of the tokenizer.
Common works are dropped into one token. Uncommon words, like syzygy, or compound words like fafasdfadsf

+ Add message

<|im_start|>user<|im_sep|>

This is a test of the tokenizer. Common works are dropped into one token. Uncommon words, like syzygy, or compound words like milkshake show up as several tokens.

<|im_end|><|im_start|>assistant<|im_sep|>

Token analysis

Inspect how your prompt is tokenized.

Token count

48

Show whitespace

<|im_start|>user<|im_sep|>

This is a test of the tokenizer. Common works are dropped into one token. Uncommon words, like syzygy, or compound words like milkshake show up as several tokens.

<|im_end|><|im_start|>assistant<|im_sep|>

200264, 1428, 200266, 279, 2500, 382, 261, 174
6, 328, 290, 99665, 13, 220, 14782, 5882, 553,
22664, 1511, 1001, 6602, 13, 220, 1367, 10269,
220, 6391, 11, 1299, 6157, 3705, 4544, 11, 503,
36762, 6391, 1299, 20018, 52038, 2356, 869, 47
2, 4919, 20290, 364, 200265, 200264, 173781, 20
0266

Prompt builder

Build and edit your conversation messages.

+ Add message

strawberry

how many letters in strawberry

Token analysis

Inspect how your prompt is tokenized.

Token count

9

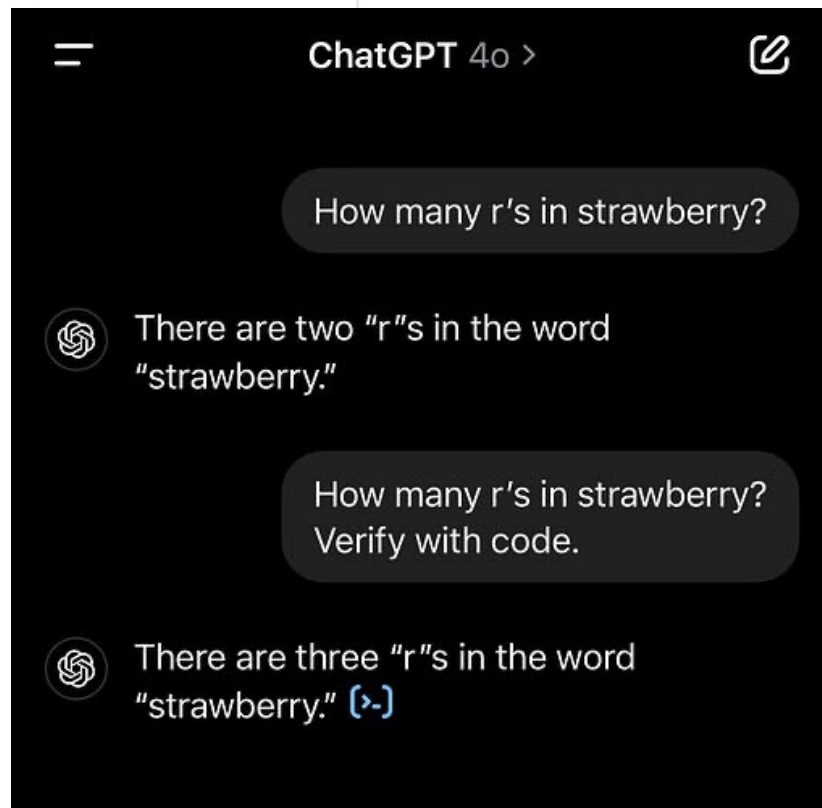
Show whitespace

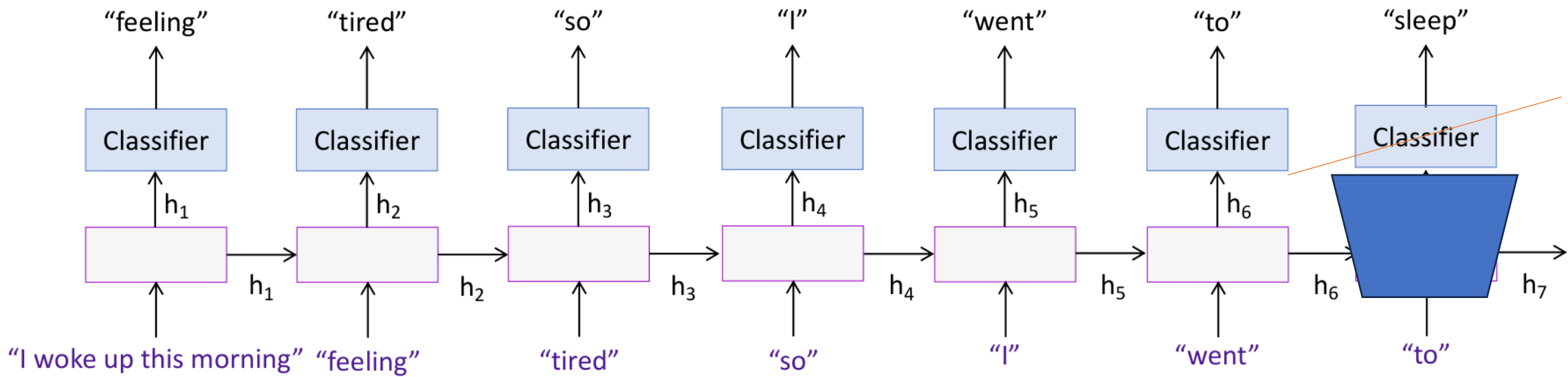


strawberry

how many letters in strawberry

In GPT tokenizers, the leading space gets glued onto the front of the following word. So strawberry and strawberry are, to the tokenizer, *two different strings*





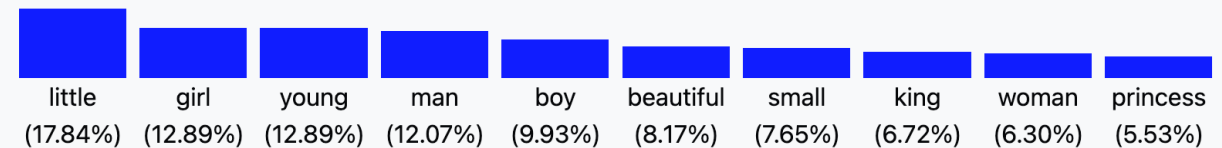
Temperature, Top-K and Top-P visualization

Parameters

Prompt

Temperature 1.9 Top-K 10 Top-P 1.00

Next Token Probabilities



<https://andreban.github.io/temperature-topk-visualizer/>

Token prediction based on training on all the internet.

Good:

- A broad, flexible base of knowledge and language.

Bad:

- Frozen: it only knows the web up to its training cutoff.
 - Blurry: common facts are sharp; rare ones are half-remembered and confabulated.
 - Unattributed: it compressed the web but didn't keep sources.
-
- Now, modern tools “Google” to answer your question for you, so they can know new facts, but those facts are “different” in it's internal representation.

ChatGPT

- Modern tools put a lot of infrastructure around this “next word prediction”
- But the general idea holds – take all of the text that humans have ever created, and learn the probability of words and phrases occurring with each other.
- Given a new input, predict (based on the probabilities you’ve learned) what the most likely next phrase/response is.
- Improved by Reinforcement Learning: Make the prediction and then let a human provide feedback on how good it is. Helps to learn “better” answers, but *implemented* as predict the next word better, and sometimes “check to see if the answer is problematic and restart”.

Two different models wear the same face

Base model. just autocompletes documents. Ask it a question and it might reply with more questions. it's imitating the web, not helping you.

Supervised fine-tuning (SFT): trained on example conversations, so it plays the role of “helpful assistant.”

+ RLHF humans rank answers; a reward model turns those rankings into a signal the model optimizes.

RLHF to make the helpful assistant is at the end

- “Helpful assistant” is a transformation base model's raw tendencies.
- Jailbreaks and Prompt Injections are untrusted text that the model treats as instructions
- So “safety” lives largely in a fine-tuning layer, not in the model's understanding.

Now it's even worse!

- Instead of answering immediately, the model writes a private chain of thought, then answers.
- Trained with reinforcement learning on problems that can be auto-graded (math, code) “verifiable rewards.”
- It gets rewarded for reaching the right answer, and discovers that thinking longer helps.
- This is the o1 / DeepSeek-R1 shift — the biggest recent change.

“try harder”

Accuracy depends on training or testing compute time.

- Old news: a bigger model (more training compute). New lever: let it think longer at answer time (test-time compute).
- Accuracy climbs with BOTH — the right-hand curve is the new idea.

openai.com/index/learning-to-reason-with-llms/

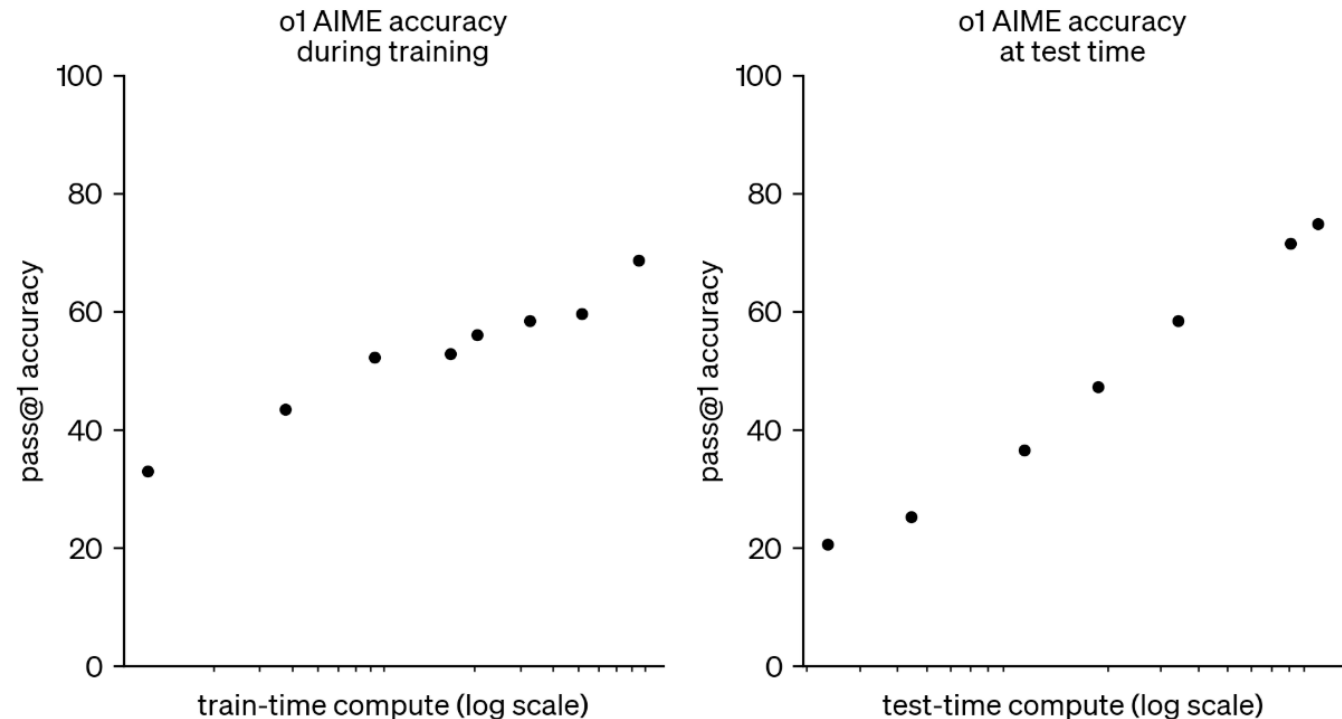
Products

Business

Developers

Company

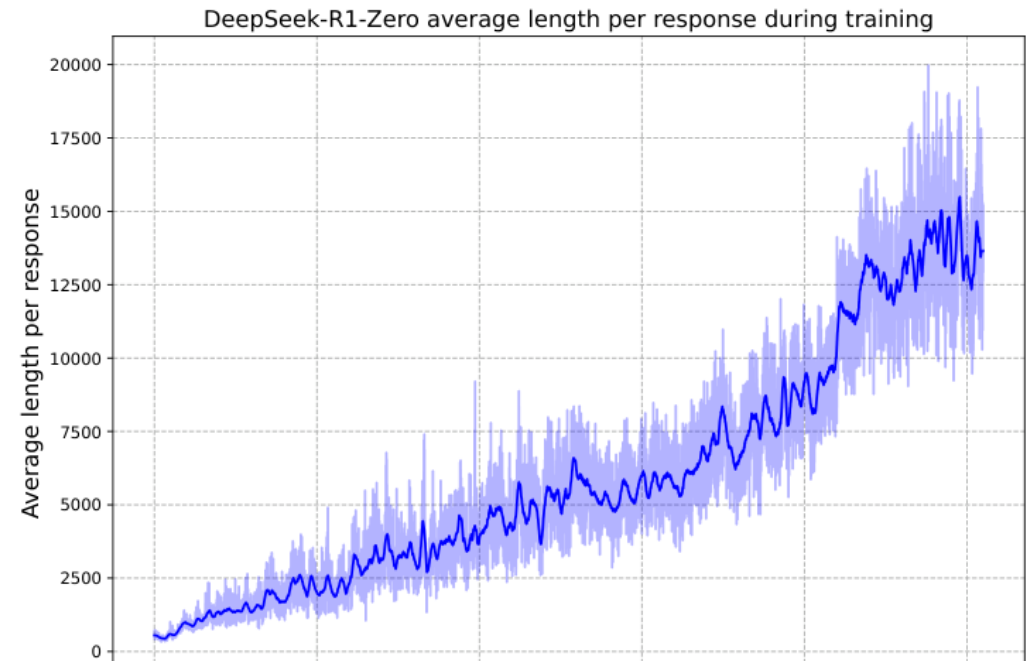
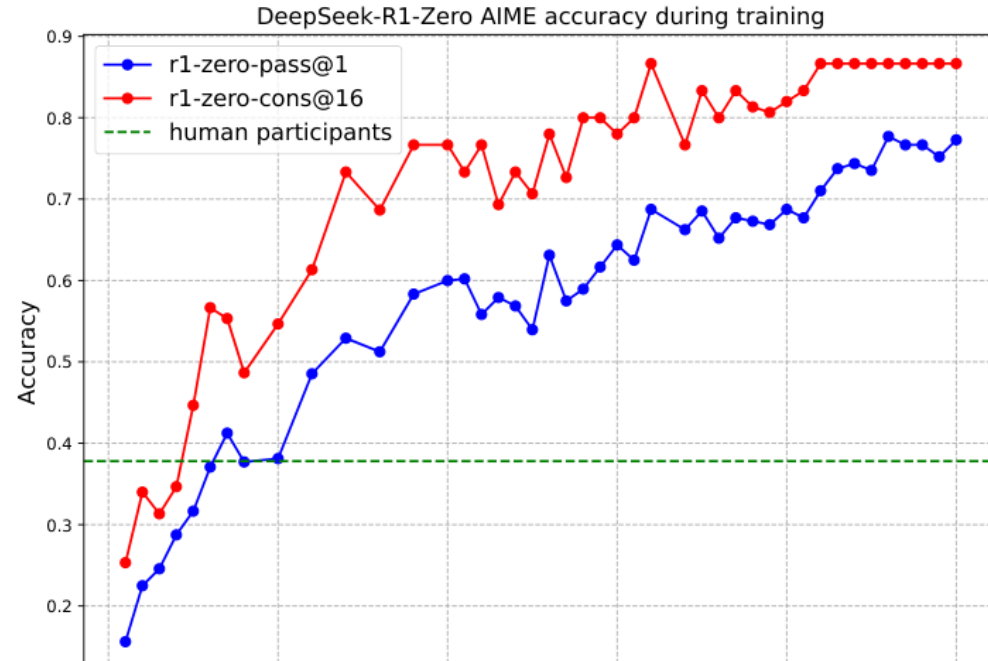
Foundation



o1 performance smoothly improves with both train-time and test-time compute

It taught itself to think longer — nobody told it to

- During RL training, DeepSeek-R1's answers grew steadily longer on their own.



DeepSeek-R1: average response length grows over RL training steps.